

E. Lucky Queries

time limit per test: 3 seconds
memory limit per test: 256 megabytes

Petya loves lucky numbers very much. Everybody knows that lucky numbers are positive integers whose decimal record contains only the lucky digits **4** and **7**. For example, numbers **47**, **744**, **4** are lucky and **5**, **17**, **467** are not.

Petya brought home string s with the length of n . The string only consists of lucky digits. The digits are numbered from the left to the right starting with 1 . Now Petya should execute m queries of the following form:

- `switch  $l$   $r$` — "switch" digits (i.e. replace them with their opposites) at all positions with indexes from l to r , inclusive: each digit **4** is replaced with **7** and each digit **7** is replaced with **4** ($1 \leq l \leq r \leq n$);
- `count` — find and print on the screen the length of the longest non-decreasing subsequence of string s .

Subsequence of a string s is a string that can be obtained from s by removing zero or more of its elements. A string is called non-decreasing if each successive digit is not less than the previous one.

Help Petya process the requests.

Input

The first line contains two integers n and m ($1 \leq n \leq 10^6$, $1 \leq m \leq 3 \cdot 10^5$) — the length of the string s and the number of queries correspondingly. The second line contains n lucky digits without spaces — Petya's initial string. Next m lines contain queries in the form described in the statement.

Output

For each query `count` print an answer on a single line.

Examples

input	Copy
2 3 47 count switch 1 2 count	
output	Copy
2 1	

input	Copy
3 5 747 count switch 1 1 count switch 1 3 count	
output	Copy
2 3 2	

Note

In the first sample the chronology of string s after some operations are fulfilled is as follows (the sought maximum subsequence is marked with **bold**):

- 47**
- 74
- 74

In the second sample:

- 747**
- 447
- 447**
- 774
- 774**

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Round 104 (Div. 1)

Finished

Practice

→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
362335204	Feb/11/2026 01:00	Accepted

→ Problem tags

data structures *2400

No tag edit access

→ Contest materials

Codeforces Round #104

Разбор задач (ru)

Tutorial #2